Modulus Functions & the Factor/Remainder Theorem

Section 1: Solving Modulus Equations

1 Solve the equation $|2x - 5| = x + 1$.

a. $x = 6$ and $x = \frac{4}{3}$

b. $x = 6$ only

c. $x = \frac{4}{3}$ only

d. $x = -6$ and $x = -\frac{4}{3}$

2 Solve $|3x + 2| = |x - 4|$.

a. $x = \frac{1}{2}$ and $x = -3$

b. $x = 2$ and $x = -3$

c. $x = -\frac{1}{2}$ and $x = 3$

d. $x = 1$ and $x = -4$

3 Why can $|x - 7| = -3$ have no real solutions?

a. The absolute value of any real expression is non-negative ($|u| \geq 0$), so it cannot equal a negative number

b. Because x must be greater than 7

c. Because -3 is not a prime number

d. The equation has complex solutions only

Section 2: Modulus Inequalities

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4 Solve the inequality $|2x - 3| < 7$.

a. $-2 < x < 5$

b. $x < 5$

c. $x < -2$ or $x > 5$

d. $-5 < x < 5$

5 Solve $|3x + 1| \geq 10$.

a. $x \leq -\frac{11}{3}$ or $x \geq 3$

b. $-\frac{11}{3} \leq x \leq 3$

c. $x \geq 3$

d. $x \leq -3$ or $x \geq \frac{11}{3}$

6 Which interval notation represents the solution to $|x - a| \leq \varepsilon$ where $\varepsilon > 0$?

a. $[a - \varepsilon, a + \varepsilon]$

b. $(a - \varepsilon, a + \varepsilon)$

c. $[-\varepsilon, \varepsilon]$

d. $[a, a + \varepsilon]$

Section 3: Polynomial Remainder & Factor Theorems

7 When $P(x) = 2x^3 - 3x^2 + ax - 5$ is divided by $(x - 2)$, the remainder is 11. Find a.

a. 6

b. 4

c. -6

d. 8

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8 If $(x+3)$ is a factor of $Q(x) = x^3 + kx^2 - 7x + 6$, find k.

a. 2

b. -2

c. 3

d. -1

9 A polynomial $f(x)$ leaves remainder 3 when divided by $(x-1)$ and remainder 7 when divided by $(x-3)$. What is the linear remainder when $f(x)$ is divided by $(x-1)(x-3)$?

a. $2x+1$

b. $2x-1$

c. $x+2$

d. $4x-1$

Section 4: Transformations of Modulus Graphs

10 How is the graph of $y = |f(x)|$ obtained from the graph of $y = f(x)$?

a. Any part of the graph lying below the x-axis is reflected vertically across the x-axis into the upper half-plane

b. The left half of the graph is discarded and replaced by reflection of the right half

c. The graph is shifted 1 unit upward

d. The graph is reflected across the y-axis

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11 How is the graph of $y = f(|x|)$ obtained from $y = f(x)$?

a. Discard the portion for $x < 0$, and reflect the portion for $x \geq 0$ across the y-axis to form an even function

b. Reflect the entire graph across the x-axis

c. Reflect the graph across $y = x$

d. Shift the graph to the right by $|x|$

12 What is the vertex (sharp turn) of the function $g(x) = 3|x - 4| - 5$?

a. $(4, -5)$

b. $(-4, -5)$

c. $(4, 5)$

d. $(0, -5)$

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